Moncada 2020 PDAC — analysis summary

GO / NO-GO GATES

Gate 1 — mixing & continuity → GO

Both patients intermix on one continuous manifold — no per-patient islands. Single connected component (radius-graph 0 detached; kNN-graph in Harmony-PCA also 1 component); iLISI median 1.00→1.97 of 2. Density dips mid-axis but no empty bins — one manifold, not two islands.

Gate 2 — module geography → GO (non-branching)

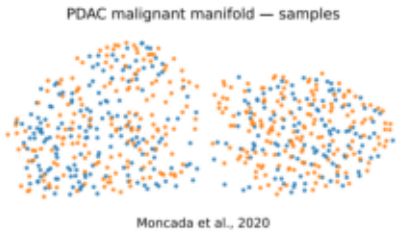
Single pSom→Gast/Crest axis: pSom high at the origin falls along pseudotime (ρ=−0.89) while Gast (ρ=+0.77) and Crest (ρ=+0.27) rise to the terminus. Gast↔pSom strongly anti-correlated in space (r=−0.89) — organised cross-repression — but the manifold does not bifurcate (0 forks). Joint pSom&Crest enrichment 0.29 (<1: inverse gradient). FLAG (passes anyway): Crest enriches the Gast end (r(Crest,Gast)=+0.38 vs r(Crest,pSom)=−0.56) — inversion of the canonical arm assignment.

Gate 3 — cross-repression → NA

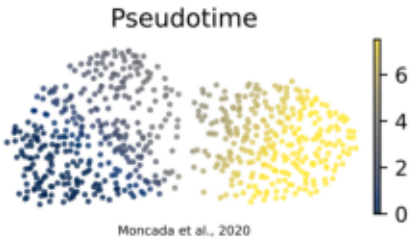
Not applicable — the manifold is non-branching (principal curve, 2 tips / 0 forks). No bifurcation, so the branching-only hub cross-repression test is not run.

STEP FIGURES

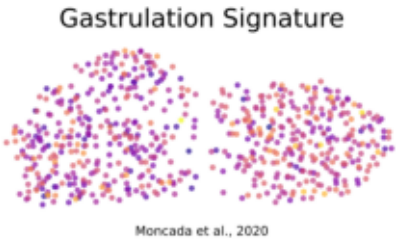
Manifold — samples



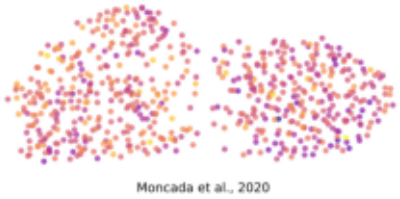
Pseudotime



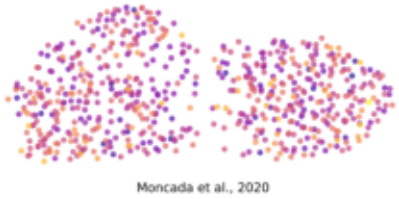
Gastrulation



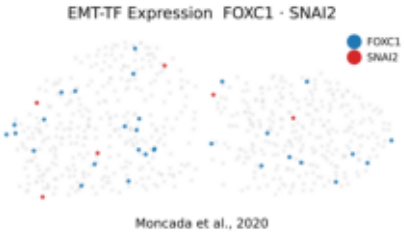
Somitogenesis



Neural Crest



EMT-TF expression



Dataset.

PDAC (Moncada et al., 2020), inDrop — 2 patients / 3,659 cells; malignant only: 635 cells (publication cell_type=="Malignant"), both samples clear the ≥100-cell floor, whole malignant manifold (no lobe scoping). Non-branching Linear-pSom/Crest.

Verdict.

Pattern reproduces in the non-branching form (Linear-pSom/Crest): one continuous somitogenesis → gastrulation/neural-crest axis, rooted at the pSom-high origin. Gast/pSom show organised cross-repression (r=−0.89) without bifurcation, so Gate 3 has no hub to test (NA). Crest inverts onto the Gast end (flagged). EMT-TFs are root-enriched (FOXC1 dominant at 46%, SNAI2 next); no distinct terminus factor. G1 GO / G2 GO / G3 NA.

Parameter choices

Sample threshold	≥100-cell floor (only 2 samples → gap-finding degenerate); both kept → 635 cells	**Overdispersed genes**	68 (scFates find_overdispersed)
Integration	Harmony on sample θ=10 (mixing reaches 2-sample chance floor)	**Embedding**	n_pcs=20, n_neighbors=30, spread=1.3 (inDrop scale); 0 islands removed
Geometry	non-branching curve N=20, λ=0.3, μ=100; 2 tips / 0 forks (2.84)	**Root**	tip 2 = Gast/pSom-high, Crest-low developmental origin
Module means (raw)	Gast −0.011 · pSom +0.115 · Crest −0.064 (Linear-pSom/Crest)	**EMT-TFs**	FOXC1 (root, 46%) & SNAI2 (root, 9%); all expressed EMT-TFs root-enriched